4. $2,3,\frac{1}{3}$ 5. 10 6. $6,4,-1$ 7. $\sum \frac{\alpha}{\beta\gamma} = \frac{2ac-b^2}{ad}$

8. $2x^2-3x-20=0$ 11. $x^3+x-12=0$

Exercise 3.2

1. When $k<0$, the polynomial has real roots.
   When $k=0$ or $k=8$, the roots are real and equal.
   When $0<k<8$ the roots are imaginary.
   When $k>8$ the roots are real and distinct.
2. $x^2-4x+7=0$ 3. $x^2-6x+13=0$ 4. x^4-16x^2+4

Exercise 3.3

1. $-3,3,\frac{1}{2}$ 2 . $\frac{2}{3},\frac{4}{3},2$ 3. $\frac{2}{3},2,6$ 4. $k=2,\ 2,\frac{1\pm\sqrt{3}}{2}$

5. $1-2i,1+2i,\sqrt{3},-\sqrt{3},\frac{1+\sqrt{37}}{2},\frac{1-\sqrt{37}}{2}$ 6. (i) $1,\frac{1}{2},3$ (ii) $-1,\frac{1}{2},\frac{3}{4}$ 7. $\pm 3,\pm\sqrt{5}$

Exercise 3.4

1. (i) $\{-2,3,-7,8\}$ (ii) $3,3,3+\sqrt{17},3-\sqrt{17}$ 2. $\left\{1,-2,\frac{-1+2\sqrt{5}}{2},\frac{-1-2\sqrt{5}}{2}\right\}$

Exercise 3.5

1. (i) $n\pi+(-1)^n\frac{\pi}{2}, n\in\mathbb{Z}$, no solution for $\sin x=4$ (ii) $2,-\frac{1}{4},\frac{2}{3}$

2. (i) $x=1$ (ii) no rational roots 3. 4^n 4. $\frac{b^2}{4a},\frac{9a^3}{b^2}$

5. (i) $2,3,\frac{1}{2},\frac{1}{3}$ (ii) $+1,-1,\frac{-3+\sqrt{5}}{2},\frac{-3-\sqrt{5}}{2}$

6. $2,3$ 7. $\frac{1}{3},3,-\frac{1}{2}$ and -2

Exercise 3.6

1. It has atmost four positive roots and atmost three negative roots.
2. It has atmost two positive roots and no negative roots.
4. It has one positive real root and one negative real root.
5. no positive real roots and no negative real roots.

Exercise 3.7

1	2	3	4	5	6	7	8	9	10
(4)	(1)	(3)	(1)	(3)	(4)	(1)	(3)	(1)	(2)

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